Quiz I

1. In a RAM, the accumulator is:
   a) The first register r_0
   b) A special output tape
   c) The program counter
   d) The location counter

2. The RAM model assumes that each register can hold:
   a) Only single-bit values
   b) Integers of arbitrary size
   c) Fixed-size integers (e.g., 32-bit)
   d) Only memory addresses

3. Under the logarithmic cost criterion, the cost of ***LOAD a*** is:
   a) $l(a) + l(0)$
   b) $l(c(0)) + t(a)$
   c) $t(a)$
   d) $l(a) + l(c(a))$

4. A RASP differs from a RAM primarily in that:
   a) It has no accumulator
   b) Its program is stored in memory
   c) It uses only indirect addressing
   d) It has no input/output tapes

5. In a RASP, each instruction occupies:
   a) 1 memory register
   b) 2 consecutive memory registers
   c) 3 consecutive memory registers
   d) A variable number of registers

6. The main advantage of the straight-line program abstraction is:
   a) It allows recursion
   b) It eliminates branching and loops
   c) It uses logarithmic cost
   d) It includes indirect addressing

7. In the bitwise computation model, variables are:
   a) Integers of arbitrary size
   b) Restricted to 0 or 1
   c) Real numbers
   d) Character strings

8. In RAM, the instruction ***STORE*** $*$ ***i*** means:
   a) Store accumulator in register i
   b) Store accumulator in register $c(i)$
   c) Store i in accumulator
   d) Store $c(i)$ in accumulator k

9. In the Turing Machine model, what does the time complexity T(n) measure?
   a) Number of arithmetic operations
   b) Number of tape head moves
   c) Number of registers used
   d) Number of comparisons

10. In the RAM model, what does the logarithmic cost of the instruction ***STORE *i*** include?
   a) Only $l(i)$
   b) $l\big(c(0)\big) + l(i) + l\big(c(i)\big)$
   c) Only $l(c(0))$
   d) $l(i) + l(c(i))$